BOUNDARY BELYI COVERS AND NORMAL-JET OBSTRUCTIONS IN THE PLANE JACOBIAN PROBLEM

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Abstract. Newton-face differential equations are a classical tool in the plane Jacobian problem. We isolate two structures carried by a face and its formal normal neighborhood.

First, a terminal face equation

$$npq - mzpq' + nzp'q = 1$$

defines a Belyi map. When the polynomial lattice has gap g, the relevant map is not the ambient degree-mnb cover but its degree-mnb/g quotient. We give its passport, prove automatic connectedness, and compute the first five quotient Hurwitz problems arising from the complete-chain queue: they have $1, 1, 1, 2, 2$ classes. Exact formulas through maximum degree 128 are infinitesimally rigid modulo source scaling.

Second, we derive the universal linear operator governing every normal layer of the determinant equation. In relative variables it is a rank-one logarithmic connection plus multiplication by the boundary-volume form. An exact formal change of normal coordinate linearizes the full nonlinear determinant equation and decouples all normal orders on one smooth boundary component. The nonlinear difficulty moves into a triangular Newton-support map. For the full $(8, 28)$ windows, the layer maps are injective from the resonant layer onward, with cokernel dimension $r - 1$ for $5 \le r \le 11$. The gauge complex also gives the complementary adjoint, an exact residue realization of every finite matrix left kernel, and a filtered one-layer descent criterion. For the archived terminal specialization, an exact replay reconstructs every layer matrix through order 8 from this operator, identifies every compatibility functional with its principal-part residue, and regenerates the fifteen normalized obstruction equations coefficient by coefficient. A normalized adjacent contact of type (a, e), with $e > a^2$, forces the explicit degree-e Belyi map

$$W_{a,e}(s) = \frac{(s-1)^{e-a^2} H_{a,e}(s)^a}{s^e}, \qquad H_{a,e} = \frac{1}{e}\, {}_2F_1\left(-a, 1; 1 - \frac{e}{a}; s\right).$$

Its passport is $(a^a, e - a^2), (e), (a + 1, 1^{e-a-1})$. The explicit finite computations use exact arithmetic and are supplied with reproducible certificates. In particular, the displayed six-polynomial terminal system is empty in characteristic zero by a complete toric special-fiber calculation. These results organize a terminal boundary-gluing program; they do not prove the plane Jacobian conjecture.

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Working computer-assisted manuscript. The recovered exact certificates and the exact-normal-form regression check have been rerun. Independent implementation and specialist review remain submission requirements.

1. Introduction

The counterexample to the Jacobian conjecture in dimension three leaves the two-variable conjecture open. The separation is geometric: the marked-root construction that fills an affine threefold does not directly fill an affine plane. In the plane, the unresolved information accumulates at infinity in Newton faces, divisorial valuations, and compatibility between successive boundary layers.

We credit Akhil Mathew as the primary human source of the problem: Mathew prompted Levent Alpöge, Alpöge prompted Fable, Fable produced the three-dimensional example, and Alpöge announced it publicly [10, 1].

The background here is substantial. One-variable face differential equations of the form used below occur in the work of Formanek and Stothers on the plane Jacobian problem [5]. Borisov developed Belyi maps and dessins in compactification frameworks for hypothetical plane counterexamples [3]. Guccione, Guccione, Horruitiner, and Valqui constructed complete-chain algorithms and reduced the range below maximum degree 125 to the pair $(72, 108)$ and its transpose [6, 7]. Normal-boundary determinant expansions also have a recent valuative precursor in Bado's work [2].

While this manuscript was being prepared, a MathOverflow answer announced a computer-assisted proof that a planar counterexample must have maximum coordinate degree at least 125 [8]. A separate exact-arithmetic campaign by Santibáñez-Leal states strong certificate evidence at the same frontier while distinguishing that evidence from a complete all-coefficient proof [9]. We do not claim the degree-125 bound. The purpose of this paper is to extract reusable boundary theorems from an independent terminal calculation and to state exactly what remains missing from a global argument.

1.1. Main results.

The first result corrects a tempting overcount. A terminal face may be written in a fractional uniformizing variable z, but the exponent lattice of the original polynomial pair can force both face polynomials to belong to $k[z^g]$. The actual Hurwitz problem is then a quotient by $u = z^g$.

Theorem 1.1 (Lattice-gap quotient, informal form)**.** *Let $m, n > 1$ be coprime, let $b \geq 1$, and suppose a terminal face satisfies*

$$npq - mzpq' + nzp'q = 1, \qquad \deg p = mb - 1, \quad \deg q = nb.$$

If $p(z) = \bar{p}(z^g)$ and $q(z) = \bar{q}(z^g)$, as forced by a complete-chain lattice gap g, then $g \mid n$ and

$$\bar{\tau}(u) = u^{n/g}\frac{\bar{p}(u)^n}{\bar{q}(u)^m}$$

is a Belyi map of degree mnb/g, with passport

$$\left(n^{(mb-1)/g}, \frac{n}{g}\right), \qquad \left(m^{nb/g}\right), \qquad \left(\frac{(m+n)b-1}{g}, 1^{\frac{mnb-(m+n)b+1}{g}}\right).$$

Every permutation triple with this passport is transitive.

Applying the theorem to the first five terminal complete-chain rows gives quotient degrees $6, 10, 9, 9, 16$ and respectively $1, 1, 1, 2, 2$ dessin classes. The first row is especially useful: an ambient degree-30 passport has eleven classes, but the gap-five lattice selects the unique cyclic pullback of

$$\frac{u(u-1)^5}{(u^2 - \frac{5}{3}u + \frac{5}{9})^3}.$$

The second group of results describes the normal direction. At layer r, new coefficients enter the determinant equation through a universal first-order operator

$$\mathscr{D}_r^{\alpha,\beta}(a,b) = (\alpha - r)a\,dB_0 - \beta B_0\,da + \alpha A_0\,db + (r-\beta)b\,dA_0.$$

Its formal adjoint is again explicit and converts coefficient obstructions into residue functionals. The same operator occurs at every normal order, with lower orders appearing in the forcing term in the original coordinates.

Writing $S = \alpha + \beta$, relative variations turn the operator into

$$\mathscr{D}_r(a,b) = A_0B_0\left(d + \frac{r}{S}\,d\log(A_0B_0)\right)\eta + \frac{S-r}{S}\xi\left(\alpha A_0\,dB_0 - \beta B_0\,dA_0\right).$$

Off resonance, provided the boundary-volume form is nonzero, it is surjective over the function field, so all obstructions come from finite Newton and pole windows. At $r = S$ it becomes an exact differential and its cokernel is controlled by ordinary residues. The full filtered-adjoint statement is proved in appendix A.

More strongly, put $M_0 = A_0B_0$ and make the formally invertible change

$$W = \frac{M_0}{AB}, \qquad T = sW^{1/S}, \qquad H = \frac{\alpha\log(B/B_0) - \beta\log(A/A_0)}{S}.$$

The complete determinant equation becomes

$$(S d + d\log M_0\,T\partial_T)\,H = (\alpha\,d\log B_0 - \beta\,d\log A_0)\left(1 - \frac{1}{S}T\partial_T\right)(W-1).$$

Thus its normal orders decouple exactly before Newton-support conditions are imposed. The inverse change is triangular on finite jets, so the nonlinearity is transferred to a canonical triangular support map rather than discarded. For the full $(8, 28)$ windows we also prove that $\mathscr{D}_r^{2,3}$ is injective for every $r \geq 5$, and hence

$$\dim\operatorname{coker}\mathscr{D}_r^{2,3} = r - 1 \qquad (5 \leq r \leq 11).$$

See appendix B.

For the stored $(8, 28)$ terminal specialization this interpretation is also verified at the artifact level in the original layer coordinates. At layers $6, 7, 8,$

the ordinary cokernel dimensions are $5, 6, 7$, while the nonlinear forcing has respectively $4, 5, 6$ nonzero compatibility pairings. After the recorded parameter normalization and removal of duplicates, the distinct equations occur with weights

$$1,\ 3,\ 5,\ 6 \qquad \text{at layers} \qquad 5,\ 6,\ 7,\ 8,$$

for a total of fifteen. Each is a scalar multiple of $\operatorname{Res}_0(\lambda(-\Phi_r))$ for an explicit left-null principal part λ, and the resulting list agrees coefficientwise with the archived terminal equations. This does not assert that the same coordinates preserve the sparse support conditions after the exact (H, W, T) change; that conjugation remains part of the support problem.

The third result gives the adjacent reduced cover forced by a normalized contact.

Theorem 1.2 (Secondary transport, informal form)**.** *For integers $a \geq 1$ and $e > a^2$, set $\delta = e - a^2$. The normalized boundary-volume transport equation has the unique polynomial-numerator solution*

$$h(s) = \frac{H_{a,e}(s)}{(s-1)^a}, \qquad H_{a,e}(s) = \sum_{k=0}^{a} \frac{a^k a!}{(a-k)! \prod_{j=0}^{k}(e-aj)} s^k.$$

The exceptional ratio

$$W_{a,e}(s) = \frac{(s-1)^\delta H_{a,e}(s)^a}{s^e}$$

is a degree-e Belyi map with passport

$$(a^a, \delta), \qquad (e), \qquad (a+1, 1^{e-a-1}).$$

For $(a, e) = (4, 17)$, this recovers the degree-17 secondary cover in the terminal $(8, 28)$ calculation. The lesson is negative but useful: reduced boundary covers are often forced and mutually compatible. Any contradiction must live in the global normal-jet gluing, not merely in the first cover.

1.2. **Scope.** The statements proved here are algebraic theorems about face equations, lattice quotients, normal-layer operators, and the listed finite Hurwitz problems. We do not prove:

- that every terminal complete chain has a complete finite-dimensional boundary deformation model;
- that the exact single-component formal solutions satisfy every sparse Newton-support and boundary-chart matching condition;
- that a support-compatible jet which lowers complexity belongs to the filtered group generated by allowable polynomial approximate roots; or
- the plane Jacobian conjecture.

Those distinctions will remain visible throughout the paper.

2. Face equations and their Belyi maps

We work over a characteristic-zero field k, embedded in $\mathbb{C}$ when using topological language about covers. A passport of a degree-D Belyi map is the ordered triple of partitions of D recording ramification over the three branch values. The order of the branch values is immaterial for our applications.

2.1. **The ambient terminal face.** Let $m, n > 1$ be coprime and $b \geq 1$. Consider polynomials

$$p, q \in k[z], \qquad \deg p = mb - 1, \quad \deg q = nb,$$

satisfying

$$npq - mzpq' + nzp'q = 1. \tag{2.1}$$

Define

$$\tau(z) = z^n \frac{p(z)^n}{q(z)^m}. \tag{2.2}$$

Proposition 2.1 (Face-to-passport formula). *The rational function τ is a degree-mnb Belyi map and*

$$\tau'(z) = z^{n-1} \frac{p(z)^{n-1}}{q(z)^{m+1}}. \tag{2.3}$$

Its passport is

$$\left(n^{mb}\right), \qquad \left(m^{nb}\right), \qquad \left((m+n)b - 1, 1^{mnb-(m+n)b+1}\right). \tag{2.4}$$

Proof. Logarithmic differentiation of (2.2) gives

$$\tau' = z^{n-1} \frac{p^{n-1}}{q^{m+1}} (npq + nzp'q - mzpq').$$

Equation (2.1) yields (2.3).

At $z = 0$, the face equation gives $np(0)q(0) = 1$. At a zero z_0 of p, it gives $nz_0 p'(z_0)q(z_0) = 1$; hence the root is simple, nonzero, and not a root of q. The analogous assertion follows for roots of q. Thus the zero fiber of τ has one point of multiplicity n at zero and $mb - 1$ further points of multiplicity n. The pole fiber consists of nb points of multiplicity m.

The numerator and denominator of τ are coprime and both have degree mnb, so $\tau(\infty) \in k^\times$. From (2.3),

$$\tau'(z) = O\left(z^{-(m+n)b}\right) \quad \text{as } z \longrightarrow \infty.$$

Therefore

$$\tau(z) - \tau(\infty) = O\left(z^{-((m+n)b-1)}\right),$$

with nonzero leading coefficient. The ramification contributions already identified total

$$mb(n-1) + nb(m-1) + ((m+n)b - 2) = 2mnb - 2.$$

Riemann–Hurwitz leaves no additional critical point. The rest of the $\tau(\infty)$-fiber is unramified, proving (2.4). $\square$

Remark 2.2. Equation (2.1) is a member of the classical family of one-variable equations studied in the Formanek–Stothers approach [5]. The content of proposition 2.1 is the complete passport forced by the stated endpoint degrees and normalization, not the first appearance of such an equation in the Jacobian problem.

2.2. **The lattice quotient.** In a complete-chain chart, z can be a fractional uniformizing coordinate. The polynomial exponent lattice forces a gap

$$g = \frac{\rho}{\gcd(\rho, \ell)}$$

and restricts the face polynomials to $k[z^g]$. The following theorem is stated purely algebraically; the complete-chain theory supplies its hypothesis in the applications.

Theorem 2.3 (Lattice-gap terminal cover)**.** *Assume the hypotheses of proposition 2.1 and suppose*

$$p(z) = \bar{p}(u), \qquad q(z) = \bar{q}(u), \qquad u = z^g,$$

for an integer $g \geq 1$. Then

$$g \mid mb - 1, \qquad g \mid nb, \qquad \gcd(g, b) = 1, \qquad g \mid n.$$

Put

$$N = \frac{n}{g}, \qquad A = \frac{mb-1}{g}, \qquad B = \frac{nb}{g},$$
$$D = \frac{mnb}{g}, \qquad H = \frac{(m+n)b-1}{g}.$$

The quotient face equation is

$$N\bar{p}\bar{q} - mu\bar{p}\bar{q}' + nu\bar{p}'\bar{q} = \frac{1}{g}, \tag{2.5}$$

and

$$\bar{\tau}(u) = u^N \frac{\bar{p}(u)^n}{\bar{q}(u)^m} \tag{2.6}$$

is a degree-D Belyi map with

$$\bar{\tau}'(u) = \frac{1}{g} u^{N-1} \frac{\bar{p}(u)^{n-1}}{\bar{q}(u)^{m+1}} \tag{2.7}$$

and passport

$$\left(n^A, N\right), \qquad \left(m^B\right), \qquad \left(H, 1^{D-H}\right). \tag{2.8}$$

Moreover,

$$\tau(z) = \bar{\tau}(z^g),$$

and every permutation triple with passport (2.8) *is transitive.*

Proof. The degree assumptions imply $g \mid mb - 1$ and $g \mid nb$. The first divisibility gives $\gcd(g, b) = 1$, so the second gives $g \mid n$. Substitution into (2.1) and division by g yield (2.5); logarithmic differentiation gives (2.7). The proof of proposition 2.1, with the zero multiplicity N at $u = 0$, gives the degree and passport. The factorization of τ is immediate.

It remains to prove transitivity. Let $(\sigma_0, \sigma_1, \sigma_\infty)$ have the stated cycle types and product one. An orbit disjoint from the unique H-cycle of σ_∞ is pointwise fixed by σ_∞. On that orbit, $\sigma_1 = \sigma_0^{-1}$, so its σ_0- and σ_1-cycles have equal length. The possible lengths for σ_0 are n and N; the length for σ_1 is m. The equality $n = m$ contradicts coprimality, while $N = m$ would imply $m \mid n$. Hence no second orbit exists. □

3. The first quotient Hurwitz problems

The complete-chain algorithm of [6, 7] supplies a finite queue of terminal data at each degree bound. We apply theorem 2.3 to the first five rows at and beyond maximum degree 125.

case	terminal $A; k$	(m, n)	g	quotient	quotient passport	classes
F_2, 125	$(7/5, 2); 1$	$(3, 5)$	5	6	$(5, 1), (3^2), (3, 1^3)$	1
one-step, 126	$(19/7, 5); 2$	$(2, 3)$	3	10	$(3^3, 1), (2^5), (8, 1^2)$	1
two-step, 126	$(11/6, 3); 1$	$(3, 2)$	2	9	$(2^4, 1), (3^3), (7, 1^2)$	1
F_{24}, 128	$(19/8, 3); 1$	$(3, 4)$	4	9	$(4^2, 1), (3^3), (5, 1^4)$	2
one-step, 132	$(19/4, 8); 1$	$(2, 3)$	3	16	$(3^5, 1), (2^8), (13, 1^3)$	2

Table 1. The first lattice-compatible terminal Hurwitz problems.

Proposition 3.1 (Exact Hurwitz count). *The final column of table 1 is exact. In every row the deck group of every connected cover is trivial, so the weighted Hurwitz count equals the ordinary number of isomorphism classes.*

Proof. For conjugacy classes $C_0, C_1, C_\infty \subseteq S_D$, the Frobenius character formula gives the weighted number

$$\frac{|C_0||C_1||C_\infty|}{(D!)^2} \sum_{\chi \in \mathrm{Irr}(S_D)} \frac{\chi(C_0)\chi(C_1)\chi(C_\infty)}{\chi(1)}.$$

The exact verifier evaluates the irreducible characters by the Murnaghan–Nakayama rule. The five resulting rational numbers are $1, 1, 1, 2, 2$.

By theorem 2.3, every triple is transitive. In all five rows $N = 1$, so the first branch permutation has a unique fixed sheet. A deck transformation centralizing the transitive monodromy group must preserve that sheet and is therefore the identity. Thus there is no automorphism weight to remove. □

3.1. **The degree-125 quotient.** For the first row, the ambient passport

$$(5^6), \qquad (3^{10}), \qquad (15, 1^{15})$$

has eleven connected dessin classes: eight with trivial deck group, two with deck group C_3, and one with deck group C_5. The gap is five, so only the last class descends to the polynomial lattice.

Proposition 3.2. *Up to source and target normalization, the unique lattice-compatible quotient is obtained from*

$$\bar{p} = 1 - u, \qquad \bar{q} = \frac{1}{5} - \frac{3}{5}u + \frac{9}{25}u^2.$$

It satisfies

$$\bar{p}\bar{q} - 3u\bar{p}\bar{q}' + 5u\bar{p}'\bar{q} = \frac{1}{5},$$

and its Belyi map is

$$\bar{\tau}(u) \doteq \frac{u(u-1)^5}{\left(u^2 - \frac{5}{3}u + \frac{5}{9}\right)^3}. \tag{3.1}$$

The ambient face is the cyclic pullback $\bar{\tau}(z^5)$.

Proof. Substitution proves the displayed differential equation. The derivative identity and passport follow from theorem 2.3. The exact character and quotient-orbit enumeration proves uniqueness. □

3.2. **Explicit post-125 covers.** For the one-step degree-126 row, put

$$P = u^3 + u^2 + \frac{5}{12}u + \frac{1}{18},$$
$$Q = u^5 + \frac{3}{2}u^4 + u^3 + \frac{1}{3}u^2 + \frac{5}{96}u + \frac{1}{576}.$$

The exact identity

$$uP^3 - Q^2 = -\frac{36u^2 + 28u + 9}{2985984} \tag{3.2}$$

gives $\bar{p} = 18P$, $\bar{q} = 192Q$.

For the two-step degree-126 row, one may take

$$\bar{p} = 1 + \frac{20}{3}u + 24u^2 + \frac{288}{7}u^3 + \frac{288}{7}u^4,$$
$$\bar{q} = \frac{1}{2} + 5u + 12u^2 + 18u^3.$$

Equivalently,

$$P = u^4 + u^3 + \frac{7}{12}u^2 + \frac{35}{216}u + \frac{7}{288}, \qquad Q = u^3 + \frac{2}{3}u^2 + \frac{5}{18}u + \frac{1}{36}$$

satisfy

$$uP^2 - Q^3 = -\frac{72u^2 + 39u + 16}{746496}. \tag{3.3}$$

For F_{24} there are two conjugate solutions over $\mathbb{Q}(\sqrt{6})$. With $\varepsilon = \pm 1$,

$$\bar{p}_\varepsilon = 1 + u + \left(\frac{1}{3} + \varepsilon\frac{\sqrt{6}}{18}\right)u^2,$$

$$\bar{q}_\varepsilon = \frac{1}{4} + \frac{5}{8}u + \left(\frac{2}{5} + \varepsilon\frac{\sqrt{6}}{40}\right)u^2 + \left(\frac{17}{160} + \varepsilon\frac{11\sqrt{6}}{480}\right)u^3.$$

Direct substitution verifies (2.5) in each case.

3.3. **Infinitesimal rigidity.** Fix the constant terms of a quotient solution and consider

$$\mathcal{F}(p,q) = Npq - mupq' + nup'q.$$

On coefficient variations with zero constant terms, its differential is

$$\mathscr{L}(\alpha,\beta) = N(\alpha q + p\beta) - mu(\alpha q' + p\beta') + nu(\alpha' q + p'\beta). \tag{3.4}$$

Source scaling gives the kernel vector

$$(\alpha,\beta) = (up', uq').$$

Proposition 3.3 (Rigidity of the explicit reduced covers)**.** *For the degree-*6*, degree-*10*, degree-*9*, and two conjugate degree-*9 *maps above, $\mathscr{L}$ is surjective and its kernel is exactly the source-scaling line. Hence each normalized quotient solution is a reduced isolated point after quotienting by source scaling.*

Proof. The exact coefficient matrices have the following ranks and certified nonzero maximal minors.

map	$(\deg p, \deg q)$	rank $\mathscr{L}$	nonzero maximal minor
F_2, 125	$(1,2)$	2	$-36/5$
one-step 126	$(3,5)$	7	2090188800
two-step 126	$(4,3)$	6	37791360/7
$F_{24},-$	$(2,3)$	4	$99/20 - 153\sqrt{6}/40$
$F_{24},+$	$(2,3)$	4	$99/20 + 153\sqrt{6}/40$

In every row the domain dimension is one more than the displayed rank. The vector (up', uq') is nonzero and lies in the kernel by differentiating source scaling. It therefore spans the kernel. □

Remark 3.4. Proposition 3.3 is a coefficient-level statement for the reduced cover. It does not prove that a full boundary deformation functor splits étale-locally into this Hurwitz point and an independent normal-jet problem.

4. The normal-boundary determinant complex

Let E be a smooth boundary curve with local coordinate z, and let s be a normal parameter. For positive integers α, β, consider

$$\alpha AB_z - \beta A_z B + s(A_z B_s - A_s B_z) = \Psi(z), \tag{4.1}$$

where

$$A = A_0 + \sum_{r\geq 1} s^r a_r, \qquad B = B_0 + \sum_{r\geq 1} s^r b_r.$$

The order-zero equation is

$$\alpha A_0 B_0' - \beta A_0' B_0 = \Psi.$$

Proposition 4.1 (Universal layer operator)**.** *At normal order r, the terms involving the new pair (a_r, b_r) are*

$$\boxed{\mathscr{D}_r^{\alpha,\beta}(a,b) = (\alpha - r)a\, dB_0 - \beta B_0\, da + \alpha A_0\, db + (r-\beta)b\, dA_0.} \tag{4.2}$$

Every other term at that order depends only on lower layers.

Proof. The coefficient of s^r in the first two terms of (4.1) contributes

$$\alpha(aB_0' + A_0 b') - \beta(a' B_0 + A_0' b).$$

The new part of the last term is $rA_0'b - raB_0'$. Combining these terms and multiplying by dz gives (4.2). □

Proposition 4.2 (Residue adjoint)**.** *For a meromorphic scalar λ,*

$$\boxed{(\mathscr{D}_r^{\alpha,\beta})^\vee(\lambda) = (\beta B_0\, d\lambda + (\alpha+\beta-r)\lambda\, dB_0, -\alpha A_0\, d\lambda + (r-\alpha-\beta)\lambda\, dA_0)\,.} \tag{4.3}$$

More precisely,

$$\begin{aligned} \lambda\mathscr{D}_r(a,b) - a\big((\mathscr{D}_r)^\vee\lambda\big)_1 - b\big((\mathscr{D}_r)^\vee\lambda\big)_2 \\ = d\big(-\beta\lambda B_0 a + \alpha\lambda A_0 b\big). \end{aligned} \tag{4.4}$$

Consequently every adjoint solution in a dual pole window gives a residue functional annihilating the image of $\mathscr{D}_r$.

Proof. Expand the right side of (4.4). Summing residues on a complete curve kills the exact differential. □

If the chosen rational-differential window represents the full dual of the target space under the residue pairing, the kernel of (4.3) represents the entire cokernel. This is the precise finite-window setting in which a left null vector of a monomial coefficient matrix becomes an intrinsic residue obstruction.

Proposition 4.3 (Virtual obstruction excess on $\mathbb{P}^1$)**.** *Suppose*

$$\mathscr{D}_r : H^0(\mathbb{P}^1, \mathcal{O}(d_A)) \oplus H^0(\mathbb{P}^1, \mathcal{O}(d_B)) \longrightarrow H^0(\mathbb{P}^1, \mathcal{O}(d_W)),$$

with $d_A, d_B, d_W \geq -1$. The target-minus-domain dimension is

$$\epsilon = d_W - d_A - d_B - 1. \tag{4.5}$$

If $\mathscr{D}_r$ is injective, its cokernel has dimension ϵ.

Proof. Use $h^0(\mathcal{O}(d)) = d + 1$ for $d \geq -1$. □

For the independently certified full $(8, 28)$ support, the windows at $r \geq 5$ are

$$d_A = 10 - r, \qquad d_B = 15 - r, \qquad d_W = 25 - r.$$

Thus

$$\epsilon_r = r - 1.$$

Proposition B.4 proves the missing injectivity for these windows. Hence this virtual excess is the actual cokernel dimension for $5 \leq r \leq 11$. The result is not used to claim the public degree-125 theorem.

5. A universal secondary Belyi map

Let $D = \{t = 0\}$ be a boundary component with coordinate s, and put

$$c(s) = \frac{s}{s-1}.$$

Assume target coordinates have been normalized to

$$\pi = t^a c(s) + O(t^{a+1}), \qquad u = \sum_{j \geq 1} t^j h_j(s), \tag{5.1}$$

where $a \geq 1$. Suppose the boundary-volume identity has no term below order t^{a+e-1} and that its first nonzero coefficient is

$$d\pi \wedge du = -t^{a+e-1}(s-1)^{-a-2}\, ds \wedge dt + O(t^{a+e}). \tag{5.2}$$

At a lower order j, the homogeneous equation is

$$jc' h_j - ach_j' = 0. \tag{5.3}$$

Lemma 5.1 (Resonance and target shears)**.** *A nonzero rational solution of* (5.3) *exists if and only if $a \mid j$. If $j = ak$, every solution is $h_j = Cc^k$, and its leading contribution $Ct^{ak}c^k$ is removed by the target shear $u \mapsto u - C\pi^k$.*

Proof. Equation (5.3) gives $h_j = Cc^{j/a}$. Since $\operatorname{div}(c) = (0) - (1)$, the power is rational exactly when j/a is an integer. The assertion about the shear follows from (5.1). □

After removing all resonant lower terms, the coefficient of (5.2) gives

$$ec'h - ach' = -(s-1)^{-a-2}. \tag{5.4}$$

5.1. The contact degree.

Proposition 5.2 (Contact-degree formula)**.** *Suppose coprime integers m, n satisfy*

$$v_E(P) = -m, \quad v_E(Q) = -n, \qquad v_D(P) = -ma, \quad v_D(Q) = -na.$$

Choose $c, d \in \mathbb{Z}$ with $dn - cm = 1$ and set

$$\pi = P^c/Q^d, \qquad \tau = Q^m/P^n.$$

Assume that in the toric chart $x = t^{-1}$ up to a unit and that

$$dP \wedge dQ = x^\kappa\, dx \wedge dy$$

up to a nonzero s-unit. Then the first possible non-shear contact degree is

$$e = a(m+n) - \kappa - 1. \tag{5.5}$$

Proof. The Bézout relation gives $v_D(\pi) = a$ and $v_D(\tau) = 0$. Direct differentiation gives

$$d\pi \wedge d\tau = -P^{c-n-1} Q^{m-d-1}\, dP \wedge dQ.$$

The monomial factor has t-order $a(m+n+1)$, while the last wedge has order $-\kappa - 2$. On the other hand, the wedge of $\pi \sim t^a$ with $\tau - \tau|_D \sim t^e$ has order $a + e - 1$. Equating orders yields (5.5). □

5.2. **Transport and passport.**

Theorem 5.3 (Universal secondary-boundary transport)**.** *Let $a, e \in \mathbb{N}$ satisfy $e > a^2$, and put $\delta = e - a^2$. Equation* (5.4) *has a unique solution*

$$h(s) = \frac{H_{a,e}(s)}{(s-1)^a}$$

whose numerator is a polynomial of degree at most a, where

$$\begin{aligned} H_{a,e}(s) &= \sum_{k=0}^{a} \frac{a^k a!}{(a-k)! \prod_{j=0}^{k} (e - aj)} s^k && (5.6) \\ &= \frac{1}{e}\, {}_2F_1\left(-a, 1; 1 - \frac{e}{a}; s\right). && (5.7) \end{aligned}$$

It satisfies

$$\begin{gathered} as(s-1)H' + (e - a^2 s)H = 1, \qquad (5.8) \\ H(0) = \frac{1}{e}, \qquad H(1) = \frac{1}{\delta}, \end{gathered}$$

and H is squarefree.

The exceptional ratio

$$W_{a,e}(s) = \frac{(s-1)^\delta H_{a,e}(s)^a}{s^e} \tag{5.9}$$

is a degree-e Belyi map. It obeys

$$W'_{a,e}(s) = \frac{(s-1)^{\delta-1} H_{a,e}(s)^{a-1}}{s^{e+1}}, \tag{5.10}$$

and its passport is

$$(a^a, \delta), \qquad (e), \qquad (a+1, 1^{e-a-1}). \tag{5.11}$$

Proof. Substitute $h = H/(s-1)^a$ into (5.4); since $c' = -(s-1)^{-2}$, this gives (5.8). Writing $H = \sum h_k s^k$ gives

$$h_0 = \frac{1}{e}, \qquad h_k = \frac{a(a+1-k)}{e - ak} h_{k-1}.$$

The denominators are nonzero because $e > a^2$, and the recurrence gives (5.6). It also proves uniqueness. The hypergeometric notation is the same terminating recurrence. Evaluation at zero and one gives the stated endpoint values. If $H(s_0) = 0$, then (5.8) gives $as_0(s_0 - 1)H'(s_0) = 1$; hence every root is simple and avoids $0, 1$.

Logarithmic differentiation of (5.9), followed by (5.8), gives

$$\frac{W'}{W} = \frac{1}{s(s-1)H},$$

which is (5.10). The numerator and denominator of W are coprime of degree e. Over zero, the a roots of H have multiplicity a and $s = 1$ has multiplicity δ. Over infinity, $s = 0$ has multiplicity e. At $s = \infty$, (5.10) shows that $W - W(\infty)$ has order $a + 1$. Finally,

$$a(a-1) + (\delta - 1) + (e - 1) + a = 2e - 2,$$

so Riemann–Hurwitz leaves only $e - a - 1$ unramified points in the third fiber and proves (5.11). $\square$

Corollary 5.4 (The $(4, 17)$ cover)**.** *For $a = 4, e = 17$,*

$$H_{4,17}(s) = \frac{195 + 240s + 320s^2 + 512s^3 + 2048s^4}{3315},$$

$$W_{4,17}(s) = \frac{(s-1)H_{4,17}(s)^4}{s^{17}}, \qquad W'_{4,17}(s) = \frac{H_{4,17}(s)^3}{s^{18}},$$

with passport $(4^4, 1), (17), (5, 1^{12})$.

6. Filtered residues and the remaining gluing problem

The phrase "vanishing high-pole residues lowers the Newton window" has an exact linear core.

Lemma 6.1 (Filtered one-layer descent)**.** *Let $D : U \to V$ be a linear map of finite-dimensional vector spaces and let $V_{\le d} \subseteq V$ be a filtered subspace. For $\Phi \in V$, the following are equivalent:*

(i) there exists $u \in U$ such that $\Phi - Du \in V_{\le d}$;

(ii) every

$$\lambda \in (\operatorname{im} D)^{\perp} \cap (V_{\le d})^{\perp}$$

satisfies $\lambda(\Phi) = 0$.

Proof. Let $q : V \to V/\operatorname{im} D$. Condition (i) says $q(\Phi) \in q(V_{\le d})$. Finite-dimensional duality says precisely that every functional on the quotient vanishing on $q(V_{\le d})$ must also vanish on $q(\Phi)$. Pulling the functionals back to V gives (ii). $\square$

For $\mathscr{D}_r$, the relevant functionals are the pole-filtered adjoint residue classes of proposition 4.2. Thus at one fixed normal layer their vanishing is equivalent to reducing the forcing term to the smaller pole window.

This is not yet a descent theorem for polynomial Keller pairs. Theorem B.1 shows that the unrestricted determinant equation on one smooth component has no further cross-order compatibility obstruction. What remains is to solve its triangular support equations, match the solutions across boundary charts, and realize a complexity-lowering finite jet by the valuation-filtered subgroup of allowable polynomial approximate-root operations.

Question 6.2 (Filtered descent)**.** *For a terminal complete-chain boundary model, does simultaneous vanishing of all high-pole residue obstructions, together with the triangular support and chart-matching equations of appendix B, either force the boundary-gluing scheme to be empty or produce an admissible finite jet of strictly smaller complete-chain complexity?*

A positive answer, together with completeness of the finite boundary model, would permit induction on complete-chain complexity. Neither hypothesis is proved here.

Question 6.3 (Filtered finite-jet realization)**.** *For a fixed complete Newton chain, through the first complexity-dropping weight, is every support-compatible formal jet in the orbit of the allowable polynomial approximate-root subgroup?*

Ordinary local Jacobian-one jets are polynomially realizable by proposition B.6; the content of question 6.3 is the extra valuation filtration imposed by the complete chain.

Question 6.4 (Normal splitting)**.** *Near one of the rigid reduced covers of proposition 3.3, does the complete boundary deformation functor split étale-locally into the reduced Hurwitz point and a normal-jet deformation problem? If not, which cross-terms measure the failure?*

7. Exact computation and reproducibility

All finite computations in this paper were rerun with exact arithmetic. The certificate suite performs the following checks.

(1) It verifies the layer operator, the adjoint integration-by-parts identity, and the specialization $\epsilon_r = r - 1$.
(2) It verifies the logarithmic wedge identity underlying the exact normal linearization, the determinant-one diagonal jet block, the $(8, 28)$ boundary-volume identity, and the terminal window dimensions.
(3) For $1 \le a \le 8$ and $\delta \in \{1, 2, 3\}$, it verifies the hypergeometric recurrence, squarefreeness, derivative identity, degrees, passports, and Riemann–Hurwitz count for $e = a^2 + \delta$.
(4) It verifies the terminal-corner arithmetic, lattice divisibilities, quotient equations, derivative formulas, and the explicit degree-6 map.
(5) It evaluates the exact symmetric-group character sums, recovers the eleven ambient degree-30 classes, and selects the unique lattice-compatible C_5 pullback.
(6) It verifies all five quotient passports and Hurwitz counts and all explicit quotient maps through maximum degree 128.
(7) It constructs the five exact tangent matrices, proves their ranks, checks their scaling kernels, and records a nonzero maximal minor in each case.

(8) Starting from the lattice points of the two normalized Newton supports, it reconstructs the exact degree-21 lower face and every deficiency layer. For the truncated support it checks the rank-14 Macaulay obstruction; for the full support it checks the weight-four square, the vertex-saturated normalization, and the resulting fifteen equations.
(9) It reconstructs every stored layer matrix through order 8 from $\mathscr{D}_r^{2,3}$, converts every left-null row to a principal part, checks the filtered adjoint equations, and recovers every compatibility pairing as a residue.
(10) It applies the recorded parameter normalization, reproduces all fifteen archived equations coefficient by coefficient, and identifies the six terminal equations as a coordinate projection of that fifteen-component obstruction section.
(11) It verifies the five degree-21 dessin representatives by explicit permutation triples, including their passports, transitivity, pairwise inequivalence, trivial deck groups, and monodromy group A_{21}.
(12) It independently replays the two exact branchwise Nullstellensatz identities using both GMP and rational Python arithmetic.
(13) It computes mixed volume 296, exhausts all 344 proper toric faces, verifies the reduced 296-point special algebra, and checks that multiplication by the sixth obstruction has nonzero determinant.

Freshly generated JSON outputs are byte-for-byte identical to the recovered reference outputs. These scripts verify the explicit identities and finite enumerations. They do not verify that the complete-chain input exhausts every hypothetical counterexample, prove the missing gluing theorem, or establish novelty.

The compact source distribution has four corresponding directories: terminal boundary calculations; the five degree-21 faces and their exact Belyi data; independent unit-ideal replay programs; and the compact 296-point toric certificate. The large selected matrices, cofactors, and residue tables are shipped as a companion data archive. A short computation index maps every computer-assisted assertion to its files.

The exact unit-ideal identities and the 296-point toric calculation are unconditional statements about their displayed polynomial systems. Their interpretation as an exhaustive below-125 obstruction still depends on the published Newton-polygon reduction from an arbitrary hypothetical counterexample below that bound to the two normalized supports used here. The original utility `lower_face_layers.py` was not recovered, but an independent exact program now regenerates the subsequent reduction from those two supports through the terminal obstruction systems. Thus the missing utility is not a dependency of the present certificate path.

Appendix A. Gauge normal form and filtered residue adjoints

The normal-layer operator of section 4 has a simpler intrinsic form. This section separates field-level solvability from the finite Newton-window obstructions used in computation.

Write
$$A = A_0, \qquad B = B_0, \qquad S = \alpha + \beta,$$
and define the boundary one-form
$$\Omega = \alpha A\, dB - \beta B\, dA.$$
For a layer variation (a, b), put
$$u = \frac{a}{A}, \qquad v = \frac{b}{B}, \qquad \xi = u + v, \qquad \eta = \alpha v - \beta u.$$

Proposition A.1 (Gauge normal form)**.** *The universal layer operator satisfies*
$$\mathscr{D}_r(a, b) = AB\, \nabla_r \eta + \frac{S - r}{S} \xi \Omega, \qquad \nabla_r = d + \frac{r}{S}\, d \log(AB).$$
If $r \neq S$ and $\Omega \neq 0$, the operator is surjective over the function field. At the resonant layer $r = S$,
$$\mathscr{D}_S(a, b) = d(\alpha A b - \beta B a).$$

Proof. Substitute
$$u = \frac{\alpha\xi - \eta}{S}, \qquad v = \frac{\beta\xi + \eta}{S}$$
into the coordinate formula for $\mathscr{D}_r$ and collect the ξ- and η-terms. Off resonance, for a rational differential Φ, set $\eta = 0$ and
$$\xi = \frac{S}{S - r} \frac{\Phi}{\Omega}.$$
At $r = S$, the multiplier term vanishes and $AB \nabla_S \eta = d(AB\eta)$. $\square$

Thus, away from resonance, a cokernel can only be created by the finite pole and Newton windows imposed on a, b, and the target differential. It is not a local obstruction to solving the rational differential equation.

Proposition A.2 (Extended deformation complex)**.** *Allow a variation $\rho\Omega$ of the normalized right-hand side and put*
$$\widetilde{\mathscr{D}}_r(a, b, \rho) = \mathscr{D}_r(a, b) - \rho\Omega.$$
Then
$$G_r(h) = \big(\alpha A h, \beta B h, (S - r) h\big)$$
satisfies
$$\widetilde{\mathscr{D}}_r \circ G_r = 0.$$
Consequently the normal layer carries a complex
$$\mathcal{G}_r \xrightarrow{G_r} \mathcal{U}_{A,r} \oplus \mathcal{U}_{B,r} \oplus \mathcal{R}_r \xrightarrow{\widetilde{\mathscr{D}}_r} \mathcal{W}_r,$$

whose degree $-1, 0, 1$ *terms represent infinitesimal gauge, deformations modulo gauge, and obstructions.*

Proof. The normal form gives

$$\mathscr{D}_r(\alpha Ah, \beta Bh) = (S - r)h\Omega.$$

□

The same normal form clarifies the adjoint. If $M = AB$, integration by parts gives

$$\lambda\mathscr{D}_r(a, b) = d(\lambda M\eta) - M\eta\,\nabla_{S-r}\lambda + \frac{S - r}{S}\lambda\xi\Omega.$$

Thus

$$(M\nabla_r)^\dagger = -M\nabla_{S-r}.$$

The complementary-layer symmetry $r \leftrightarrow S - r$ is invisible in the original four-term coordinate formula.

Theorem A.3 (Filtered residue adjoint)**.** *Let* $U_{A,r}, U_{B,r} \subset k(E)$ *be the permitted correction windows and let* W_r *be the permitted target window. Represent* $W_r^\vee$ *by principal parts* $\lambda = (\lambda_p)$ *at the allowed poles, with pairing*

$$\langle\lambda, \omega\rangle = \sum_p \mathrm{Res}_p(\lambda_p\omega).$$

Then

$$(\operatorname{coker}\mathscr{D}_r)^\vee \cong \ker\mathscr{D}_{r,\mathrm{win}}^\vee,$$

where the filtered adjoint condition is that

$$\beta B\,d\lambda + (S - r)\lambda\,dB$$

annihilate $U_{A,r}$*, and*

$$-\alpha A\,d\lambda + (r - S)\lambda\,dA$$

annihilate $U_{B,r}$*. The equation*

$$\mathscr{D}_r(a_r, b_r) = -\Phi_r$$

is solvable exactly when

$$\sum_p \mathrm{Res}_p(\lambda_p\Phi_r) = 0$$

for every filtered adjoint class.

Proof. Multiply the coordinate operator by λ, integrate each derivative term by parts, and apply the residue theorem. Boundary terms are precisely the displayed principal-part pairings. Finite-dimensional duality then identifies the cokernel dual with the filtered adjoint kernel. □

There is an exact coefficient dictionary. For target basis

$$dz, z\,dz, \dots, z^d\,dz$$

and a left-null vector $c = (c_0, \dots, c_d)$, set

$$\lambda_c(z) = \sum_{j=0}^{d} c_j z^{-j-1}.$$

Then

$$\operatorname{Res}_{z=0}\left(\lambda_c \sum_{j=0}^{d} f_j z^j\,dz\right) = \sum_{j=0}^{d} c_j f_j.$$

Thus a matrix null vector and its principal-part representative are exactly the same functional in two coordinate systems.

Corollary A.4 (Resonant de Rham obstruction)**.** *On $E = \mathbb{P}^1$, let Σ be the pole set and suppose the allowed primitive window is complete modulo constants. At $r = S$,*

$$\operatorname{coker} \mathscr{D}_S \cong H^1_{\mathrm{dR}}(\mathbb{P}^1 \setminus \Sigma) \cong \left\{(\rho_p)_{p\in\Sigma} : \sum_p \rho_p = 0\right\}.$$

In particular, its dimension is $|\Sigma| - 1$.

Proof. At resonance the image consists of exact differentials. A rational differential on $\mathbb{P}^1$ has a rational primitive exactly when all residues vanish, and the residue theorem supplies the single relation among them. □

Status of the six terminal components. The preceding results prove the operator normal form, gauge complex, and filtered residue realization abstractly. Proposition B.4 now proves that the ordinary full-support layer maps have cokernel dimensions $r - 1$ for $5 \leq r \leq 11$. The exact artifact replay goes further in the original layer coordinates: it reconstructs the stored matrices from $\mathscr{D}_r^{2,3}$, checks every filtered adjoint equation, and writes every compatibility functional as a residue pairing with an explicit principal part. The nonzero pairing counts at layers $6, 7, 8$ are $4, 5, 6$. After the recorded normalization and duplicate removal, the fifteen distinct equations have layer counts $1, 3, 5, 6$ at $r = 5, 6, 7, 8$, and they agree coefficientwise with the archived list. The six equations used in the terminal contradiction are an explicit coordinate projection of this fifteen-component residue section.

The remaining comparison is narrower: the archived normalized equations have not been conjugated term by term into the new (H, W, T)-coordinates. Consequently we do not claim that the exact normal-coordinate change preserves the required sparse Newton windows or identifies the admissible approximate-root orbit.

Appendix B. Exact normal linearization and finite-jet realization

The layer operator of section 4 is the differential of an exact formal normal form. This removes one apparent source of difficulty: on a single smooth boundary component, before Newton-support conditions are imposed, the normal orders do not interact nonlinearly.

B.1. **An exact change of variables.** Let $K = k(E)$ be the function field of the boundary curve, equipped with the differential $d = d_E$, and work in $K[[s]]$. Retain the notation

$$S = \alpha + \beta, \qquad M_0 = A_0 B_0, \qquad \omega = \alpha\, d\log B_0 - \beta\, d\log A_0 = \frac{\Psi}{M_0}\, dz.$$

For units $A = A_0 + O(s)$ and $B = B_0 + O(s)$, define

$$W = \frac{M_0}{AB} \in 1 + sK[[s]], \tag{B.1}$$

$$T = sW^{1/S} \in s + s^2 K[[s]], \tag{B.2}$$

and

$$H = \frac{1}{S}\left[\alpha \log\left(\frac{B}{B_0}\right) - \beta \log\left(\frac{A}{A_0}\right)\right] \in sK[[s]]. \tag{B.3}$$

The fractional power and logarithms are the unique formal series with the displayed constant terms. Since $T = s + O(s^2)$, it has a unique formal inverse.

Theorem B.1 (Exact normal linearization)**.** *The transformation* (B.1)–(B.3) *is formally invertible, with*

$$s = TW^{-1/S}, \qquad A = A_0 W^{-\alpha/S} e^{-H}, \qquad B = B_0 W^{-\beta/S} e^{H}. \tag{B.4}$$

After regarding H and W as series in T, the full determinant equation (4.1) *is equivalent to the linear equation*

$$\boxed{(Sd + d\log M_0\, T\partial_T)\, H = \omega \left(1 - \frac{1}{S} T\partial_T\right)(W - 1).} \tag{B.5}$$

Proof. The inverse formulas follow by solving the two linear equations in $\log(A/A_0)$ and $\log(B/B_0)$. Put

$$P = s^{-\alpha} A, \qquad Q = s^{-\beta} B.$$

Then (B.4) gives

$$P = T^{-\alpha} A_0 e^{-H}, \qquad Q = T^{-\beta} B_0 e^{H}.$$

In the coordinates (z, T), direct logarithmic differentiation gives

$$dP \wedge dQ = T^{-S-1} M_0 \big(\alpha\, d\log B_0 - \beta\, d\log A_0 + S\, dH + d\log M_0\, T\partial_T H\big) \wedge dT. \tag{B.6}$$

On the other hand, differentiating $s = TW^{-1/S}$ at fixed z gives

$$s^{-S-1}\,dz \wedge ds = T^{-S-1}\left(W - \frac{T}{S}\partial_T W\right) dz \wedge dT. \tag{B.7}$$

Equation (4.1) is precisely

$$dP \wedge dQ = s^{-S-1}\Psi(z)\,dz \wedge ds.$$

Substitute (B.6) and (B.7), and cancel the order-zero identity

$$M_0\,(\alpha\,d\log B_0 - \beta\,d\log A_0) = \Psi(z)\,dz.$$

The result is (B.5). □

Corollary B.2 (Decoupled normal orders)**.** *Write*

$$H = \sum_{r\geq 1} h_r T^r, \qquad W = 1 + \sum_{r\geq 1} w_r T^r.$$

Then the full determinant equation is equivalent to the independent family

$$\boxed{S\left(d + \frac{r}{S}\,d\log M_0\right) h_r = \frac{S-r}{S} w_r \omega \qquad (r \geq 1).} \tag{B.8}$$

At resonance $r = S$, this becomes

$$d(M_0 h_S) = 0.$$

Consequently, over K and without sparse support restrictions, solutions at the separate normal orders assemble uniquely into one formal solution in the (H, W, T)-coordinates.

Proof. Take the coefficient of T^r in (B.5). Formal invertibility then gives the last assertion. □

The relation with proposition A.1 is exact. If

$$u_r = \frac{a_r}{A_0}, \qquad v_r = \frac{b_r}{B_0}, \qquad \xi_r = u_r + v_r, \qquad \eta_r = \alpha v_r - \beta u_r,$$

then triangularity of the change of variables gives

$$h_r = \frac{\eta_r}{S} + \text{lower-layer terms}, \qquad w_r = -\xi_r + \text{lower-layer terms}. \tag{B.9}$$

The homogeneous part of (B.8) is therefore the gauge normal form of the universal layer operator.

B.2. **The nonlinear support map.** The exact linearization does not preserve the polynomial Newton windows. Indeed, the inverse formulas give universal triangular expressions

$$\frac{a_r}{A_0} = -h_r - \frac{\alpha}{S} w_r + F_{A,r}(h_{<r}, w_{<r}), \tag{B.10}$$

$$\frac{b_r}{B_0} = h_r - \frac{\beta}{S} w_r + F_{B,r}(h_{<r}, w_{<r}), \tag{B.11}$$

where the $F_{\bullet,r}$ are polynomials in lower normal orders. If $U_{A,r}, U_{B,r} \subset K$ are the permitted coefficient windows, set

$$\begin{aligned} \kappa_r(H,W) &= (\kappa_{A,r}, \kappa_{B,r}), \\ \kappa_{A,r} &= \left[A_0\Big(-h_r - \frac{\alpha}{S}w_r + F_{A,r}\Big)\right]_{K/U_{A,r}}, \\ \kappa_{B,r} &= \left[B_0\Big(h_r - \frac{\beta}{S}w_r + F_{B,r}\Big)\right]_{K/U_{B,r}}. \end{aligned} \tag{B.12}$$

Then the original Newton-support condition is exactly $\kappa_r(H,W) = 0$ for all r.

For every finite N, the change from the first N pairs (a_r, b_r) to the first N pairs (h_r, w_r) is a triangular polynomial automorphism of jet spaces. Its diagonal block is

$$(u_r, v_r) \longmapsto \left(\frac{\alpha v_r - \beta u_r}{S}, -(u_r + v_r)\right),$$

whose determinant is 1. Thus a stored finite boundary system may be conjugated into the exact coordinates without changing its scheme. The determinant equations become linear; the nonlinear information moves into the triangular support map $\kappa = (\kappa_r)$.

Remark B.3 (What the conductor does not prove). Normalization and conductor quotients can make lattice descent finite when a semigroup is not saturated. They do not force an infinite supported formal series to terminate polynomially. In the full $(8, 28)$ chart the toric coordinate change is unimodular and the relevant high-order cross-sections are complete integer intervals. The decisive obstruction is therefore a finite-window and finite-jet realization problem, not merely a hole in a normalized semigroup.

B.3. **The full** $(8, 28)$ **layer maps.** For the degree-21 lower face of appendix C, put

$$\alpha = 2, \qquad \beta = 3, \qquad S = 5, \qquad A_0 = zp, \qquad B_0 = z^2 q.$$

The face equation (C.1) gives

$$M_0 = z^3 pq, \qquad \Omega = 2A_0\, dB_0 - 3B_0\, dA_0 = z^2\, dz, \qquad \omega = \frac{dz}{zpq}.$$

The exact equation (B.5) is

$$5H_z + T\left(\frac{3}{z} + \frac{p'}{p} + \frac{q'}{q}\right) H_T = \frac{1}{zpq}\left(W - \frac{T}{5}W_T - 1\right). \tag{B.13}$$

Proposition B.4 (Injectivity of the terminal layer maps)**.** *Assume* p, q *satisfy* (C.1)*. For* $r \geq 5$*, let*

$$a \in k[z]_{\leq 10-r}, \qquad b \in k[z]_{\leq 15-r},$$

where a negative degree bound means the zero space. Then

$$\mathscr{D}_r^{2,3}(a,b) = 0 \quad \Longrightarrow \quad a = b = 0.$$

In particular, for $5 \le r \le 11$,

$$\dim \operatorname{coker} \mathscr{D}_r^{2,3} = r - 1. \tag{B.14}$$

Proof. At $r = 5$, put

$$C = 2A_0 b - 3B_0 a.$$

By resonance, $\mathscr{D}_5(a, b) = dC$. If this vanishes, then C is constant; both terms are divisible by z, so $C = 0$. Hence

$$2pb = 3zqa.$$

Equation (C.1) implies $\gcd(p, zq) = 1$, so $p \mid a$. Since $\deg a \le 5 < 7 = \deg p$, one has $a = 0$, and then $b = 0$.

Now suppose $r > 5$, and set

$$C = 2A_0 b - 3B_0 a, \qquad E = A_0 b + B_0 a.$$

The gauge formula and $\mathscr{D}_r(a, b) = 0$ give

$$E = zpq \left(C \frac{M_0'}{M_0} + \frac{5}{r-5} C' \right). \tag{B.15}$$

If $C \neq 0$ and $d = \deg C$, the right side has degree exactly $d + 17$: its leading coefficient contains the nonzero factor

$$20 + \frac{5d}{r-5}.$$

On the other hand,

$$\deg E \le \max\{\deg A_0 + \deg b, \deg B_0 + \deg a\} \le 23 - r.$$

Thus $d \le 6 - r$. This is impossible for $r \ge 7$; for $r = 6$ it forces $d = 0$, whereas C is divisible by z. Hence $C = 0$, and the coprimality and degree argument above again gives $a = b = 0$.

For $5 \le r \le 11$, proposition 4.3 applies to the stated full-support windows and gives the target-minus-domain dimension $r - 1$. Injectivity proves (B.14). □

Remark B.5 (Comparison with the archived terminal equations)*.* For the archived terminal specialization, the exact residue replay gives ordinary cokernel dimensions $5, 6, 7$ at layers $6, 7, 8$, but the specialized nonlinear forcing has only

$$4, \quad 5, \quad 6,$$

nonzero compatibility pairings. Every pairing is recovered as $\operatorname{Res}_0(\lambda(-\Phi_r))$ for an explicit left-null principal part. After imposing the recorded normalization, the distinct surviving equations have layer counts $1, 3, 5, 6$ at $r = 5, 6, 7, 8$; these fifteen polynomials agree coefficientwise with the archived system. Thus the residue origin of the archived equations is an exact computer-assisted theorem in the original layer coordinates.

What has not been checked is a term-by-term conjugation of this normalized system into the new (H, W, T)-coordinates. That comparison concerns the triangular support map, normalization directions, and admissible approximate-root action; it is not needed for the residue reconstruction just stated.

B.4. **Ordinary finite jets are polynomially realizable.** Full algebraization of an infinite formal series is stronger than is needed to lower a lexicographically ordered boundary complexity. The first weight at which a leading datum is cancelled is determined by a finite jet. For unrestricted local area-preserving coordinates, such finite jets have no algebraization obstruction.

Proposition B.6 (Finite-jet realization)**.** *Let k be an infinite field of characteristic zero, and let $\widehat{\phi} \in \operatorname{Aut} k[[x, y]]$ fix the origin and satisfy $\det J\widehat{\phi} = 1$. For every N, there is a polynomial automorphism $\phi_N \in \operatorname{Aut} k[x, y]$, with $\det J\phi_N = 1$, whose N-jet agrees with that of $\widehat{\phi}$. It may be chosen as a finite composition of linear $\mathrm{SL}_2(k)$ maps and polynomial shears.*

Proof. Match the linear part first. Inductively, suppose the first unmatched homogeneous term has degree n:

$$(x, y) + (f_n, g_n) + O(\mathfrak{m}^{n+1}).$$

The Jacobian-one condition gives $\partial_x f_n + \partial_y g_n = 0$, so for a homogeneous binary form K_{n+1},

$$(f_n, g_n) = (\partial_y K_{n+1}, -\partial_x K_{n+1}).$$

Over an infinite field, powers of linear forms span the binary forms of each degree. Write

$$K_{n+1} = \sum_i c_i \ell_i^{n+1}, \qquad \ell_i = a_i x + b_i y.$$

The Hamiltonian flow of each summand is the exact polynomial shear

$$(x, y) \longmapsto (x, y) + c_i(n + 1)\ell_i^n (b_i, -a_i),$$

because ℓ_i is constant in the direction $(b_i, -a_i)$. Composing these shears matches the degree-n discrepancy and changes only higher degrees. Induction through degree N proves the claim. $\square$

Question B.7 (Filtered finite-jet realization)**.** *Proposition B.6 concerns the full local Jacobian-one automorphism group. The approximate-root operations allowed by a complete chain form a smaller valuation-filtered subgroup. The remaining realization problem is therefore the finite family of maps*

$$\mathrm{gr}_r \mathcal{G}_{\mathrm{approx}} \longrightarrow \mathrm{gr}_r \mathcal{G}_{\mathrm{formal}},$$

where the two groups denote admissible polynomial approximate-root operations and formal complexity-lowering operations, respectively. Surjectivity through the first complexity-dropping weight realizes the drop; a failure of surjectivity produces a canonical finite obstruction.

Appendix C. The degree-21 divisor and terminal certificates

The body develops the lattice-gap quotient covers. Here we record the complementary calculation on the original lower face and the exact certificates obtained after substituting its five dessins into the two surviving Newton supports.

C.1. **The lower face.** For either surviving support in the reduction of Guccione et al. [7], the transformed bracket is

$$[P, Q] = X^2.$$

For the primitive toric valuation

$$\nu(X) = -2, \qquad \nu(Y) = 1,$$

put $z = XY^2$. The lower faces have the form

$$P_{\text{face}} = Xp(z), \qquad Q_{\text{face}} = X^2 Y q(z),$$

where $\deg p = 7$, $\deg q = 10$, and

$$pq + 2zpq' - 3zp'q = 1. \tag{C.1}$$

Proposition C.1 (The degree-21 Belyi map)**.** *The rational function*

$$\tau(z) = z\frac{q(z)^2}{p(z)^3}$$

*is a degree-*21 *Belyi map with passport*

$$(2^{10}1), \qquad (3^7), \qquad (17\,1^4).$$

There are exactly five connected dessin isomorphism classes with this passport. They form one arithmetic orbit over an irreducible quintic field and have monodromy A_{21}.

Proof. Equation (C.1) implies that p, q are coprime and nonzero at zero. Differentiation gives

$$\tau'(z) = \frac{q(z)}{p(z)^4}.$$

Thus the ten roots of q give double zeros, the seven roots of p give triple poles, and infinity has ramification index 17. The total ramification is

$$10 + 14 + 16 = 40 = 2 \cdot 21 - 2,$$

so there is no additional branch value.

The exact Murnaghan–Nakayama character calculation in S_{21}, followed by the class-size factor, gives $5 \cdot 21!$ labeled triples. Each is transitive, and the deck group is trivial, so there are five isomorphism classes. Exact coefficient reconstruction places all five over one quintic field; exact permutation certificates give monodromy A_{21}. The enumeration and reconstruction are computer-assisted. □

C.2. **Identification with Borisov's divisor.**

Theorem C.2 (Valuation over F_{-5})**.** *Let E_ν be the source divisor defined by ν. Its target center is Borisov's divisor F_{-5}. The normal ramification index and residue degree are*

$$(e, f) = (1, 21).$$

Consequently neither surviving Newton support can realize Borisov's Three-dessin framework.

Proof. Set $t = Y$, so $x = t^2/z$ in the original source coordinates. The source volume form is

$$dx \wedge dy = t^{-6} z^2 \, dt \wedge dz.$$

After adding the reduced boundary coefficient, the augmented-canonical label is -5.

The face expansions are

$$P = t^{-2} z p(z) + \text{higher } \nu\text{-terms}, \qquad Q = t^{-3} z^2 q(z) + \text{higher } \nu\text{-terms}.$$

On the target put

$$s = \frac{P}{Q}, \qquad \lambda = \frac{Q^2}{P^3}.$$

Then s is a uniformizer and

$$dP \wedge dQ = -s^{-6} \lambda^{-3} \, ds \wedge d\lambda,$$

so the target label is also -5. Pullback gives

$$\frac{P}{Q} = t \left(\frac{p(z)}{zq(z)} + O(t) \right),$$

hence $e = 1$. The residue of λ is

$$\operatorname{res}_{E_\nu}(\lambda) = z \frac{q(z)^2}{p(z)^3} = \tau(z),$$

so $f = 21$ by proposition C.1.

In Borisov's target coordinates $(y_1, y_2) = (Q, P)$, the divisor F_{-5} is characterized by valuations $(-3, -2)$ and by the nonconstant residual parameter y_1^2/y_2^3. Those are exactly the values above. Later infinitely-near divisors with the same two numerical orders have constant residue of this parameter, so they are distinguished.

For clarity, the complete dictionary at the generic points is

	E_ν on the source	F_{-5} on the target
normal parameter	$t = Y$	$s = P/Q$
residual parameter	$z = XY^2$	$\lambda = Q^2/P^3$
orders of (P, Q)	$(-2, -3)$	$(-2, -3)$
normal map	$s = t(p/(zq) + O(t))$	$e = 1$
residue map	$z \longmapsto zq^2/p^3$	$f = 21$

Equivalently, at the completed generic points the map has the form

$$\operatorname{Spec} k(E_\nu)[[t]] \xrightarrow{s=t\,u(z,t),\ \lambda=\tau(z)+O(t)} \operatorname{Spec} k(F_{-5})[[s]],$$

where $u(z,0)=p(z)/(zq(z))$ is a unit and $[k(E_\nu):k(F_{-5})]=\deg\tau=21$.

Borisov's Three-dessin framework gives residue degree 16 on its unique source divisor above F_{-5} [4]. The forced divisor has residue degree 21, a contradiction. This excludes the named framework, not the two supports themselves. $\square$

Corollary C.3. *Any plane counterexample of maximum coordinate degree below* 125 *has generic degree at least* 21*.*

Proof. The published below-125 reduction leaves the two supports above, and the generic-degree formula contains the contribution $ef=21$ from theorem C.2. $\square$

Question C.4. *Is E_ν the only source divisor over the generic point of F_{-5}? The calculation $f=21$ is one residue-degree contribution; cancellation valuations or infinitely-near divisors could contribute additional sheets.*

C.3. **Exact support certificates.** An independent exact program starts with all lattice points of each normalized Newton support and the deficiencies

$$d_P(i,j)=j-2i+2, \qquad d_Q(i,j)=j-2i+3.$$

It reconstructs the lower face over the quintic field and checks all eighteen coefficients of (C.1). For the truncated support, seven weight-three compatibility equations and eighteen weight-four equations span all fourteen weight-four monomials in the four effective positive-weight parameters. Hence these parameters lie in the radical of the obstruction ideal, forcing the required top vertices to vanish. The vertex-saturated truncated system is therefore empty in characteristic zero. Because the computation is carried out in the quintic coefficient field, the same conclusion holds for all five conjugate dessins.

For the full support, the same direct layer recursion reaches a weight-four perfect square. On its zero set the top P- and Q-vertex coefficients are nonzero multiples of $t_{1,1}^2$ and $t_{1,1}^3$; vertex saturation therefore permits the normalization $t_{1,1}=1$. Exact layer elimination then reduces the problem to fifteen equations in five variables over the same quintic field. For the residue and five-variable terminal systems, write

$$K_0=\mathbb{Q}[u]/(u^5-u^4+3u^3+3u^2+26).$$

Six selected equations already generate the unit ideal.

Proposition C.5 (Exact residue provenance of the terminal equations)**.** *For the stored lower face and normal-layer forcing terms through order* 8*, every archived layer matrix is the matrix of $\mathscr{D}_r^{2,3}$ in the displayed monomial windows. Every left-null row is represented by an explicit principal part λ*

satisfying the filtered adjoint equations, and its compatibility polynomial is exactly

$$\mathrm{Res}_{z=0}\big(\lambda(-\Phi_r)\big).$$

Before parameter normalization, the numbers of nonzero compatibility polynomials at layers $6, 7, 8$ *are* $4, 5, 6$. *After the recorded normalization and duplicate removal, the distinct equations have counts*

$$(1, 3, 5, 6) \quad \textit{at layers} \quad (5, 6, 7, 8).$$

The resulting fifteen equations agree coefficientwise with the archived normalized system, and the six equations used below are the coordinate projection with zero-based archived indices $4, 6, 8, 9, 10, 11$.

Exact computer-assisted proof. The supplement rebuilds the lower face over $\mathbb{Q}[u]/(u^5 - u^4 + 3u^3 + 3u^2 + 26)$, forms every matrix directly from $\mathscr{D}_r^{2,3}$, and computes exact row-reduction transforms. For each left-null row it constructs the dual principal part and verifies the two filtered formal-adjoint equations coefficientwise. Pairing that principal part with $-\Phi_r$ reproduces the row-reduced compatibility polynomial exactly. The normalization map is then applied symbolically; the resulting fifteen polynomials are compared term by term with the archived JSON representation. The replay writes the bases, matrices, principal parts, normalization map, comparison, and projection matrix to machine-readable audit files. $\square$

Theorem C.6 (Five-variable terminal unit-ideal certificate)**.** *For the stored normalized full-support system, the six selected obstruction equations generate the unit ideal over* K_0.

Exact computer-assisted certificate. The Macaulay multiplication map has 10,824 columns. The certificate specifies 7,121 rows and 7,121 pivot columns. Reducing this exact minor at a good prime and at the selected quintic embedding gives

$$\det = 859 \pmod{2053}.$$

The determinant is therefore nonzero in characteristic zero. The certificate package regenerates all fifteen exact equations, rebuilds the matrix, and replays the fixed pivot minor. $\square$

There is also a later, branchwise exact certificate for the same terminal analysis. After the ordinary branches are eliminated, two exceptional systems remain; denote their ideals in $K_\eta[I, H, G, A]$ by I_+ and I_-, where $K_\eta = \mathbb{Q}[\eta]/(T(\eta))$ and the explicit quintic T is recorded with the branch certificate. We keep this field model separate from K_0; the branch certificate is checked in its displayed primitive element and does not require an unstated identification of the two.

Theorem C.7 (Branchwise exact Nullstellensatz certificates)**.** *For each sign, the stored nine-generator ideal is the unit ideal:*

$$I_+ = K_\eta[I, H, G, A], \qquad I_- = K_\eta[I, H, G, A].$$

More precisely, the supplement gives cofactors $C_{\epsilon,j}$ such that

$$\sum_{j=0}^{8} C_{\epsilon,j} F_{\epsilon,j} = 1, \qquad \epsilon \in \{+,-\},$$

and every product $C_{\epsilon,j} F_{\epsilon,j}$ has total degree at most five.

Exact computer-assisted certificate. The monomials of total degree at most five in four variables span a 126-dimensional space. Modular row reduction at $p = 31$, using a root of the quintic field polynomial, selects 111 rows. The two selected 111×111 minors have determinants 1 and 17 modulo 31. Exact rational elimination then produces 80 nonzero cofactor monomials for each sign. A separate GMP program and an independently written rational-arithmetic Python implementation multiply the cofactors by the original generators, reduce in the quintic field, and recover exactly the constant polynomial 1. □

The compact source distribution contains the branch generators and replay programs; the companion large-data archive contains the selected matrices and exact cofactors. This certificate is unconditional for the two stored ideals. Its use in the global support analysis still depends on the prior reduction from the five-variable terminal system to the two exceptional branches.

Combining the truncated resultant with either terminal certificate architecture eliminates both encoded support alternatives after the degree-21 face substitution. This statement is exact for the stored systems. Turning it into a stand-alone global below-125 proof requires a line-by-line audit that the published Newton reduction, the five faces, all saturations and normalizations, the ordinary-branch elimination, and the layer systems exhaust every candidate.

Theorem C.8 (Compact toric terminal certificate)**.** *Let*

$$\rho = F_4, \qquad (g_1, g_2, g_3, g_4, g_5) = (F_6, F_8, F_9, F_{10}, F_{11})$$

be the six exact normalized obstruction polynomials in the supplement. Then

$$V(\rho, g_1, g_2, g_3, g_4, g_5)(\overline{K_0}) = \varnothing$$

over the quintic coefficient field K_0.

Exact computer-assisted proof. At $p = 2053$ and the split value $u = 216$, the five g_i have mixed volume 296. The Minkowski sum has 344 proper faces: 270 have a monomial initial form, and exact saturated Laurent calculations show that the other 74 initial ideals are unit ideals. Hence the entire special toric intersection is the reduced 296-point scheme represented by the archived multiplication matrices. Multiplication by ρ has determinant $682 \neq 0 \pmod{2053}$.

The toric compactification is proper, so no component is lost at infinity. Localizing the coefficient ring at $(2053, u - 216)$ therefore gives a finite étale scheme of rank 296 over the unramified local DVR, and ρ remains a unit. Thus the six polynomials have no common characteristic-zero

zero. Repeating over the five split embeddings gives determinant residues $682, 116, 337, 242, 740$, whose product is $51 \pmod{2053}$. □

This theorem is unconditional for the displayed six polynomials. Its use as an obstruction to the two normalized terminal $(8, 28)$ support alternatives is connected to those raw supports by the independent exact reconstruction above. The remaining global dependency is the published Newton-polygon reduction asserting that every below-125 Keller pair is covered by those alternatives.

C.4. **A two-chart layer-seven certificate.** The large layer-eight terminal certificate is not needed for the stored degree-21 full-support specialization. A second exact calculation identifies the missing layer-four operation, passes to its adjacent Newton chart, and obtains a contradiction using only layers five through seven.

Proposition C.9 (The layer-four direction is a chart transition)**.** *The one-dimensional residual kernel at normal layer four is not induced by a non-trivial fixed-chart polynomial or Laurent source automorphism. It is the infinitesimal form of the complete-chain operation*

$$Y \longmapsto Y + \lambda X^{-4},$$

which changes the completed Newton chart.

In adjacent blowdown coordinates

$$u = (xy)^{-1}, \qquad v = y, \qquad w = v - v_*,$$

write

$$\widehat{P} = Lu + P_2 + P_3 + \cdots, \qquad \widehat{Q} = Muw + Nu^2 + Q_3 + Q_4 + \cdots.$$

The Jacobian equation forces

$$L = 0, \qquad F_5 = 0, \qquad 2P_2\partial_w Q_3 - 3Q_3\partial_w P_2 = 0,$$

and hence

$$P_2 = CR^2, \qquad Q_3 = GR^3$$

for a common linear form R.

Proof. The fixed-chart kernel is identified with weighted-divergence-free vector fields by the exact identity

$$\mathscr{D}_r\Theta_r(f, g) = (f\Psi)' + (r - 5)g\Psi, \qquad \Psi = z^2.$$

Exact support-window calculation gives residual dimensions $(1, 2, 1, 1)$ in layers one through four. A nonlinear fixed-chart shear calculation forces its leading parameter to vanish. Conjugating the $k = 4$ complete-chain operation into the adjacent chart gives the stated translation, and the first homogeneous parts of the transformed Jacobian identity give the three displayed equations. Unique factorization yields the common approximate root. □

Theorem C.10 (Stored terminal no-gluing theorem)**.** *After the canonical $k = 4$ rechart and the forced adjacent-chart condition of proposition C.9, the complete layer-five-through-seven support and chart-matching equations of the stored degree-21 terminal specialization have no common zero over the algebraic closure of*

$$K_0 = \mathbb{Q}[u]/(u^5 - u^4 + 3u^3 + 3u^2 + 26).$$

Exact computer-assisted proof. After the adjacent-chart linear equation is imposed, the layer-five equation is affine-linear in one remaining coefficient d:

$$F_5 = D(x)d + R(x, a, b).$$

On $D = 0$, the specialized layer-five-through-seven equations have an exact nineteen-term Nullstellensatz identity.

On $D \neq 0$, eliminating d reduces the problem to a five-generator ideal $J \subset K_0[x, a, b]$. At the good prime $(2053, u - 216)$, integral points are excluded by a 201-term affine unit certificate. The common normal fan has 93 face patterns: 61 have a monomial initial form, 32 require exact residue-torus tests, and the ten surviving cones are covered by two explicit weighted charts. Exact identities in the x-dominant and a-dominant charts exclude those cones. Properness of the weighted compactification lifts emptiness from the good fiber to characteristic zero. An independent weighted-projective implementation gives the same conclusion. ☐

Theorem C.10 is exact for the displayed stored terminal system and has been replayed from the recovered source package. It does not independently prove that every plane Keller pair below degree 125 reaches this system. That global interpretation still depends on the upstream Newton-polygon reduction and its exhaustiveness; neither the plane Jacobian conjecture nor a new independent proof of the 125 bound is asserted here.

C.5. **Credit for the** 125 **bound.** MathOverflow user `ratto3423` publicly announced a lower bound of 125 for the maximum coordinate degree of any characteristic-zero plane counterexample. The claim rests on a computer-assisted elimination of the same two supports [8]. We credit the public theorem announcement to `ratto3423`. The answer does not disclose enough method to compare its computation with the certificates above.

Accordingly, our claim is not priority for the degree-125 conclusion. The potential contribution of this project is the explicit degree-21 boundary description and an independently auditable terminal certificate architecture. The current certificate is conditional only at the upstream exhaustiveness layer described above, not at its final linear-algebra step.

Companion results and research register

This reader edition is accompanied by a separate *Results and Research Register*. The register preserves secondary proved results, conditional developments, open questions, corrections, and computational leads without making them part of this paper's theorem spine. Proof-bearing technical developments omitted from this reader PDF remain in the versioned source tree and in the complete version-8 archival edition. The v9 primary-home map distinguishes the paper responsible for a result from other papers that use or motivate it.

AI and computational disclosure

AI systems were used extensively in exploration, symbolic-program development, proof drafting, source organization, and manuscript editing. They are not authors. Human specialist review has not yet been completed. The evidence for the computer-assisted propositions is the exact certificate suite described above, not language-model output.

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